Practical 9:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
~
  GNU nano 8.7
int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > far) {
                        far = next;
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}
```

Output:

```
ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ nano optimal.c

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ gcc optimal.c -o optimal

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ ./optimal
Hits = 5
Faults = 7
Hit % = 41.67
Fault % = 58.33

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$
```